CO2xN: Porewater Ammonium

July 2026 Samples

2026-10-19

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##Run Information

```
cat("Run Information: ") #lets you know what section you're in
```

```
## Run Information:
```

```
library(here)
```

```
## here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW
```

```
#set the run date & user name
  sample_year <- "2026"
  sample_month <- "July"
  user <- "Brenden Blakley"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  NH3_files <- c("Raw Data/2M_July2026_NH4_Run1.csv",
                 "Raw Data/2M_July2026_NH4_Run2.csv")

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- here("Porewater QAQC Logs", "GCReW_NH4_QAQC_Log.csv")
  file_path_QAQC <- here("Porewater QAQC Logs", "GCReW_SEAL_Check_QAQC_Log.csv")
  final_path <- "Processed Data/CO2xN_Porewater_NH4_202607.csv"

#record any notes about the run or anything other info here:
  run_notes <- "Slope slightly out of range, all other checks pass - runs kept"

  #Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

  cat(run_notes)
```

```
## Slope slightly out of range, all other checks pass - runs kept
```

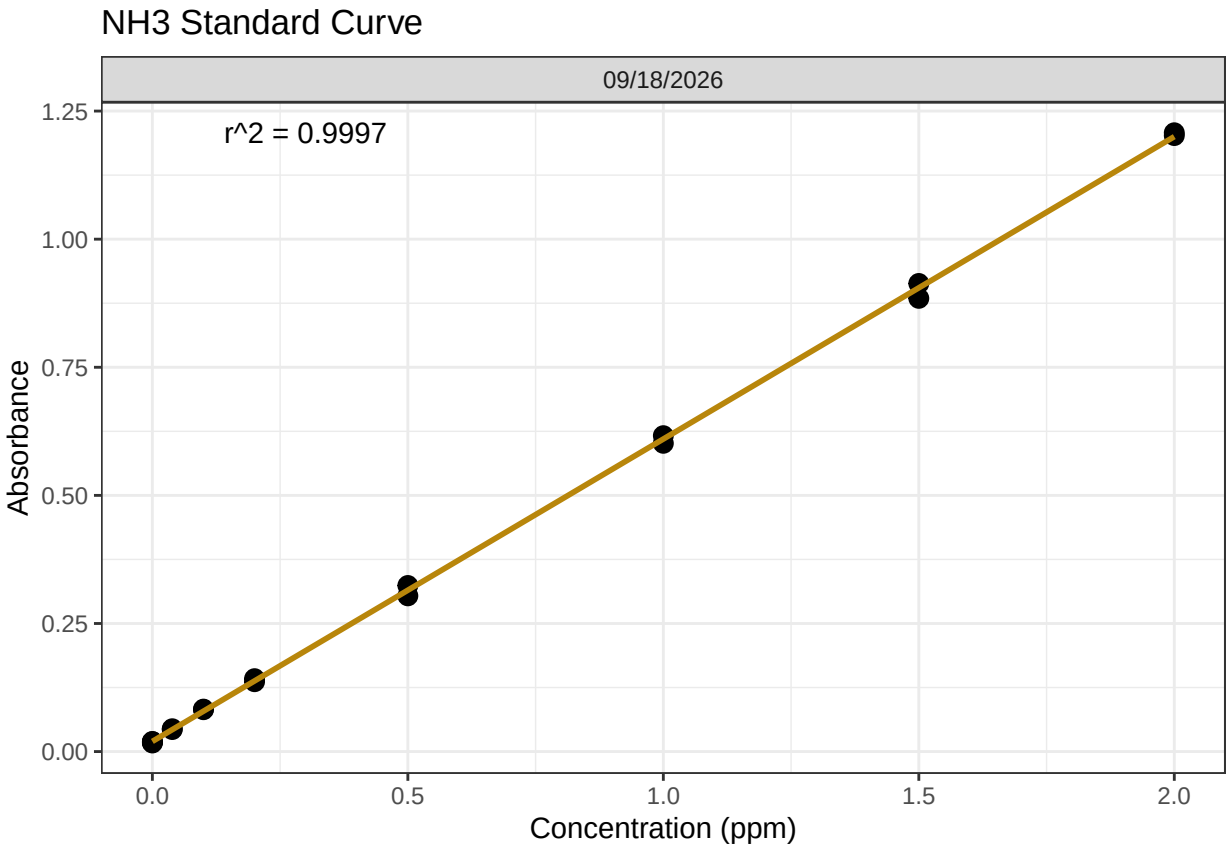
##Setup

##Read in metadata and create similar sample IDs for matching to samples

##Import Data & Clean

0.1 Assessing standard Curves

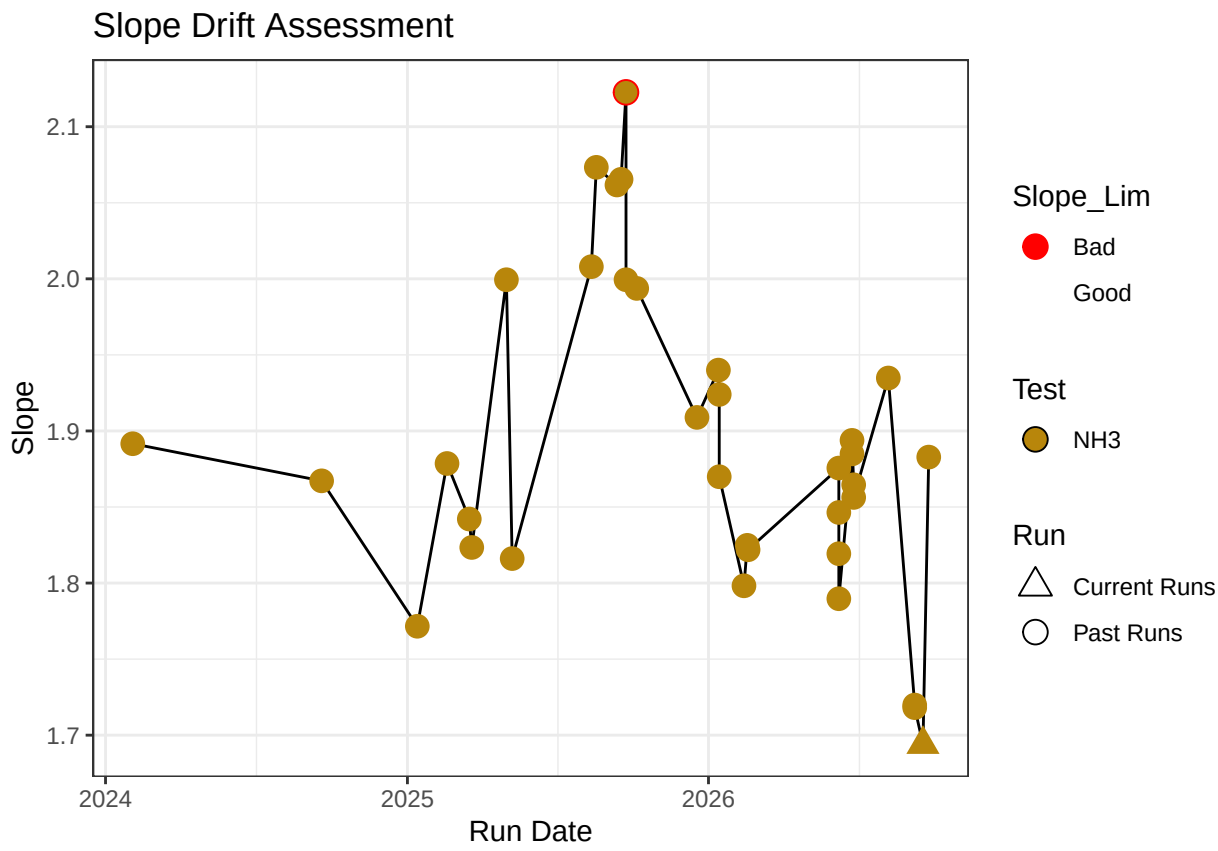
```
## `geom_smooth()` using formula = 'y ~ x'
```



Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	09/18/2026	R2_Pass	0.9996735	1.694039

0.2 Update Standard Log

```
## `summarise()` has grouped output by 'Test', 'Intercept', 'Slope', 'R_Squared',
## 'User'. You can override using the `.groups` argument.
```


Slope Drift Assessment
Slope
2.1
2.0
1.9
1.8
1.7
2024
2025
2026
Run Date
Slope_Lim
Bad
Good
Test
NH3
Run
Current Runs
Past Runs



##Write Standard Log

0.3 Dilution Corrections - ensure the latest dilution is kept

```
## [1] "Reran Samples: c(\"L1-40\", \"L1-80\", \"L4-40\", \"L4-80\", \"L5-80\", \"L10-80\", \"L11-20\",
```

```
## [1] "Diluted Samples: c(\"L1-40\", \"L1-80\", \"L4-40\", \"L4-80\", \"L5-80\", \"L10-80\", \"L11-20\
```

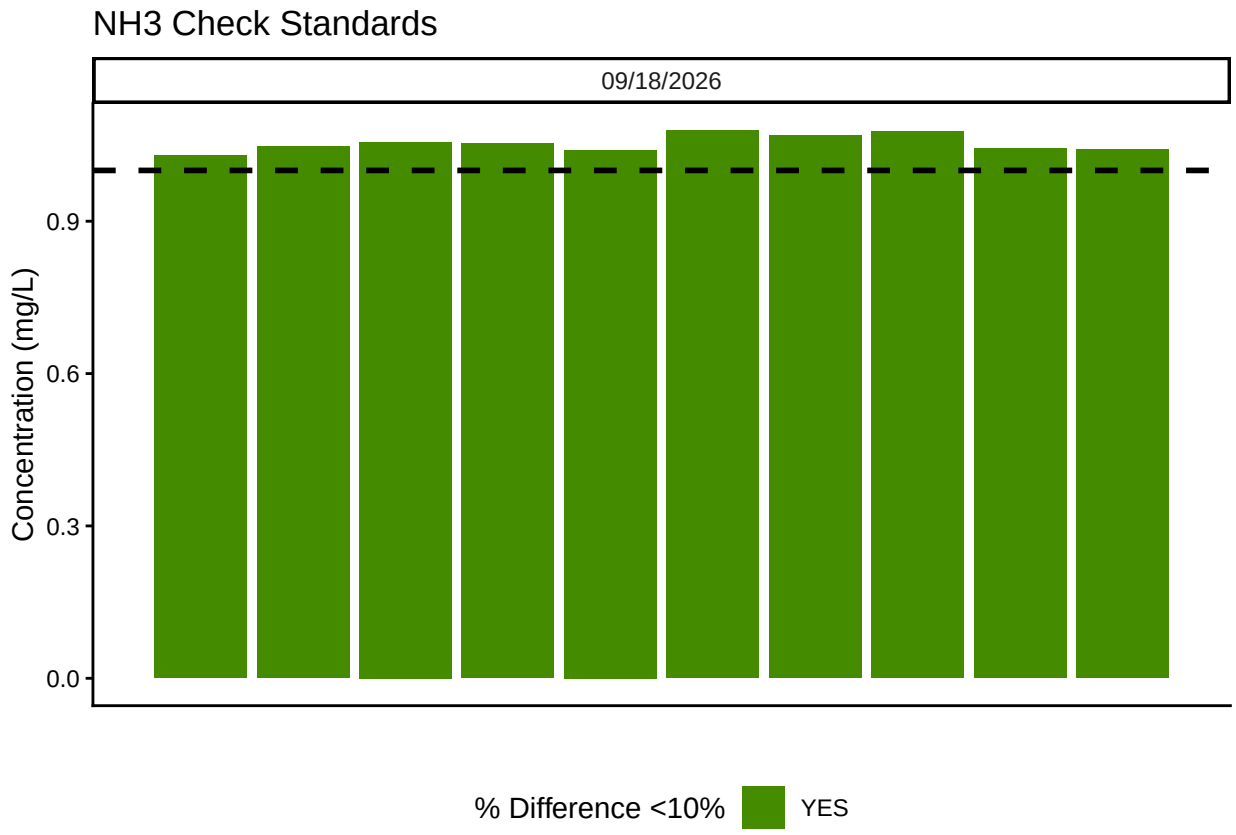
```
## [1] "No Naming Issues Detected"
```

0.4 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	09/18/2026	Performance Check Within 25% - PROCEED	1.103069	1.034
Ammonia 2	09/18/2026	Performance Check Within 25% - PROCEED	1.103069	1.034

0.5 Analyze the Check Standards

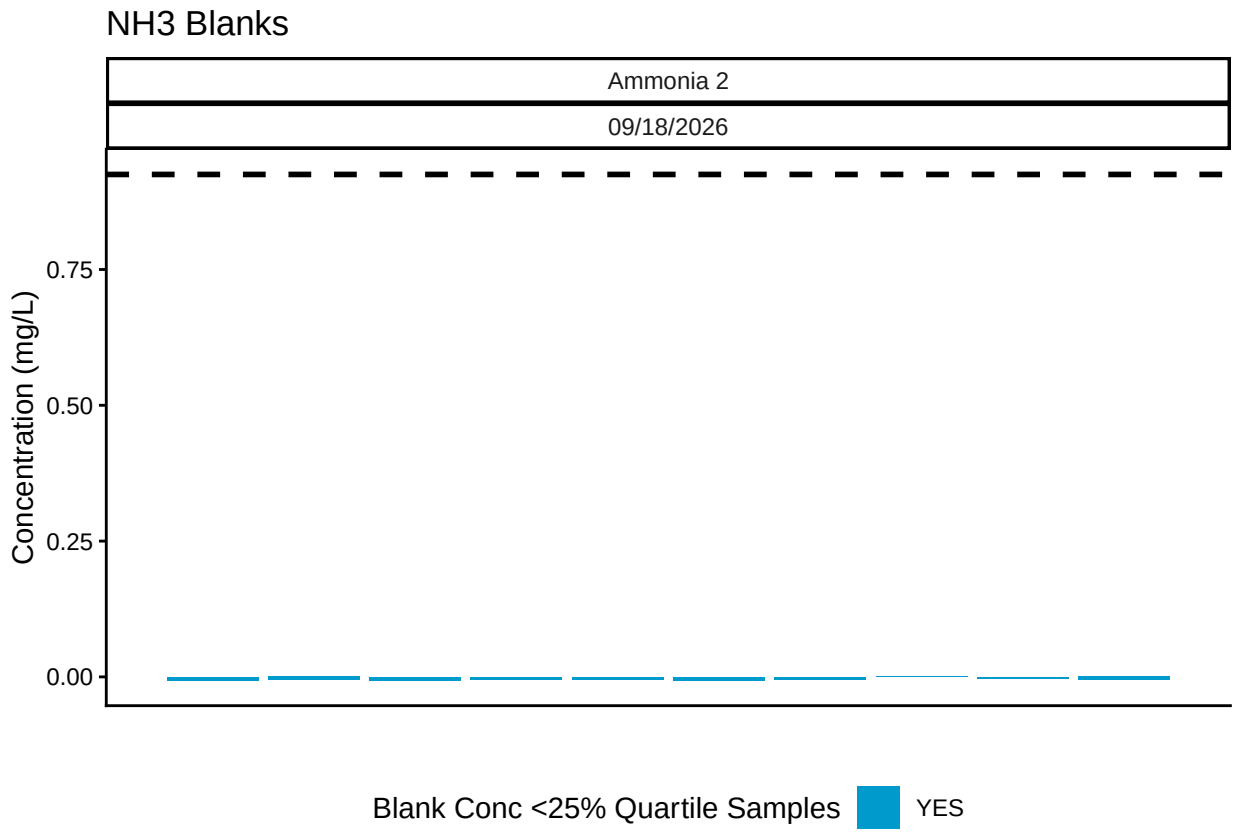
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	09/18/2026	RSV WITHIN RANGE - PROCEED	0.0156808	0.25



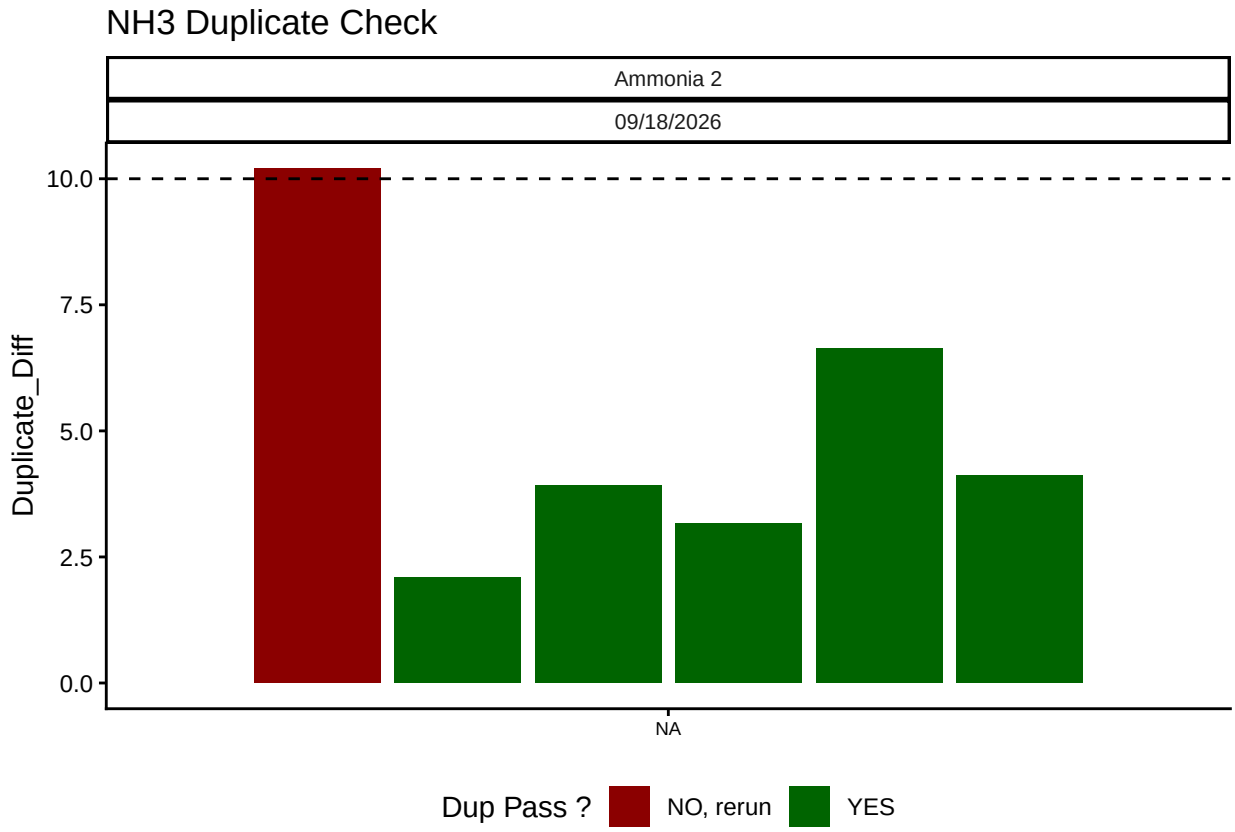
Test	Run_Date	CHK_Flag
Ammonia 2	09/18/2026	>60% of Checks Pass - PROCEED

0.6 Analyze Blanks

Test	Run_Date	BLK_Pct_Flag	Mean_Blk_Conc	Quantile_25
Ammonia 2	09/18/2026	>60% of Blanks Pass - PROCEED	-0.004594	0.925009

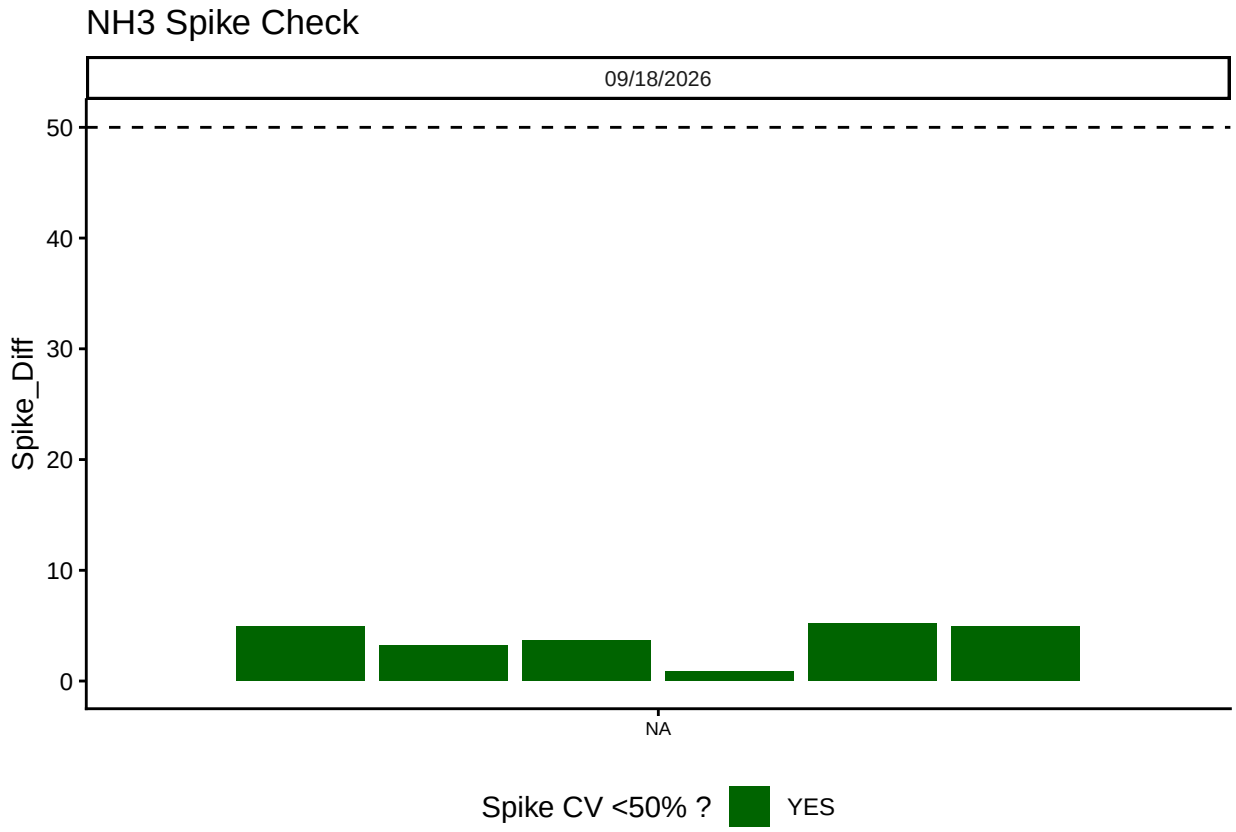


0.7 Analyze Duplicates



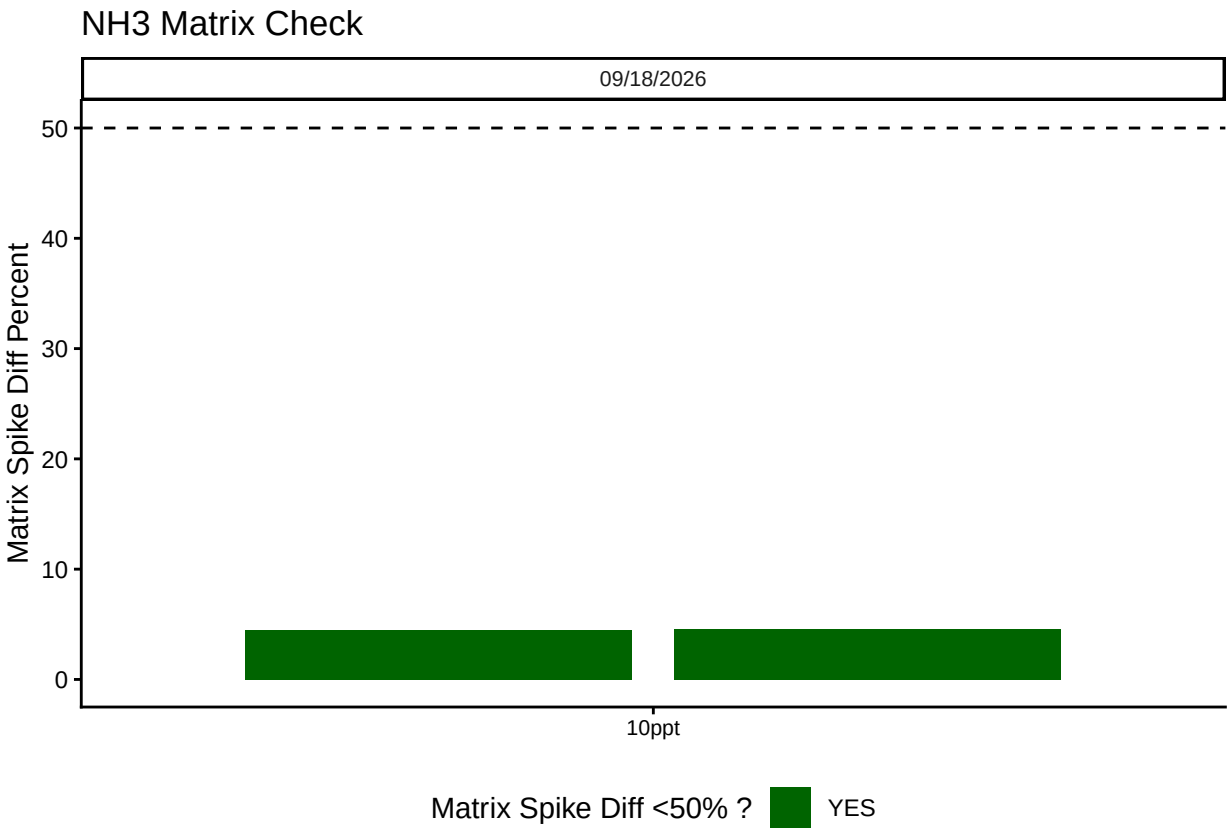
Test	Run_Date	Dup_Flags
Ammonia 2	09/18/2026	>60% of Dups Pass - PROCEED

0.8 Spikes



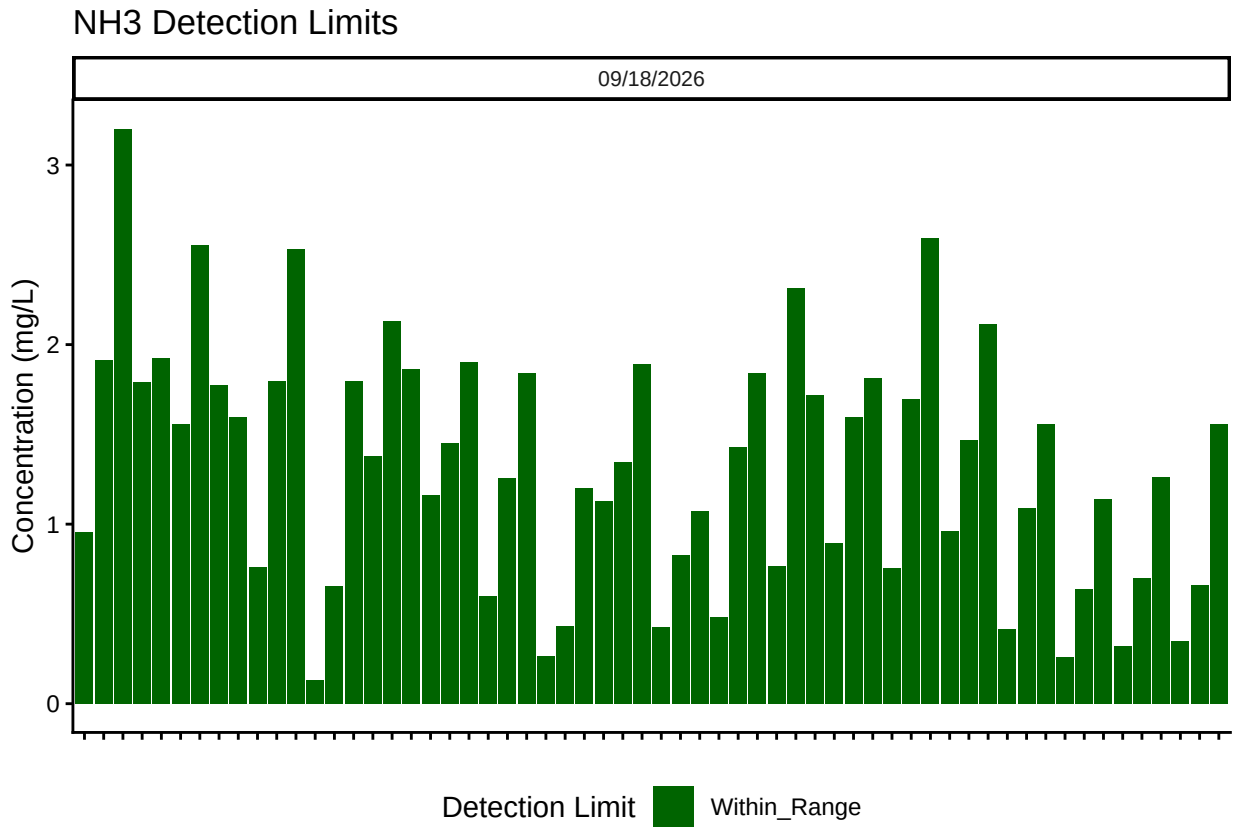
Test	Run_Date	Spike_Flags
Ammonia 2	09/18/2026	>60% of Spikes have a Diff <50% - PROCEED

0.9 Matrix Effects



Test	Run_Date	Matrix_Flags
Ammonia 2	09/18/2026	Matrix Has Diff <50% - PROCEED

##Sample Flagging - Within range of standard curve



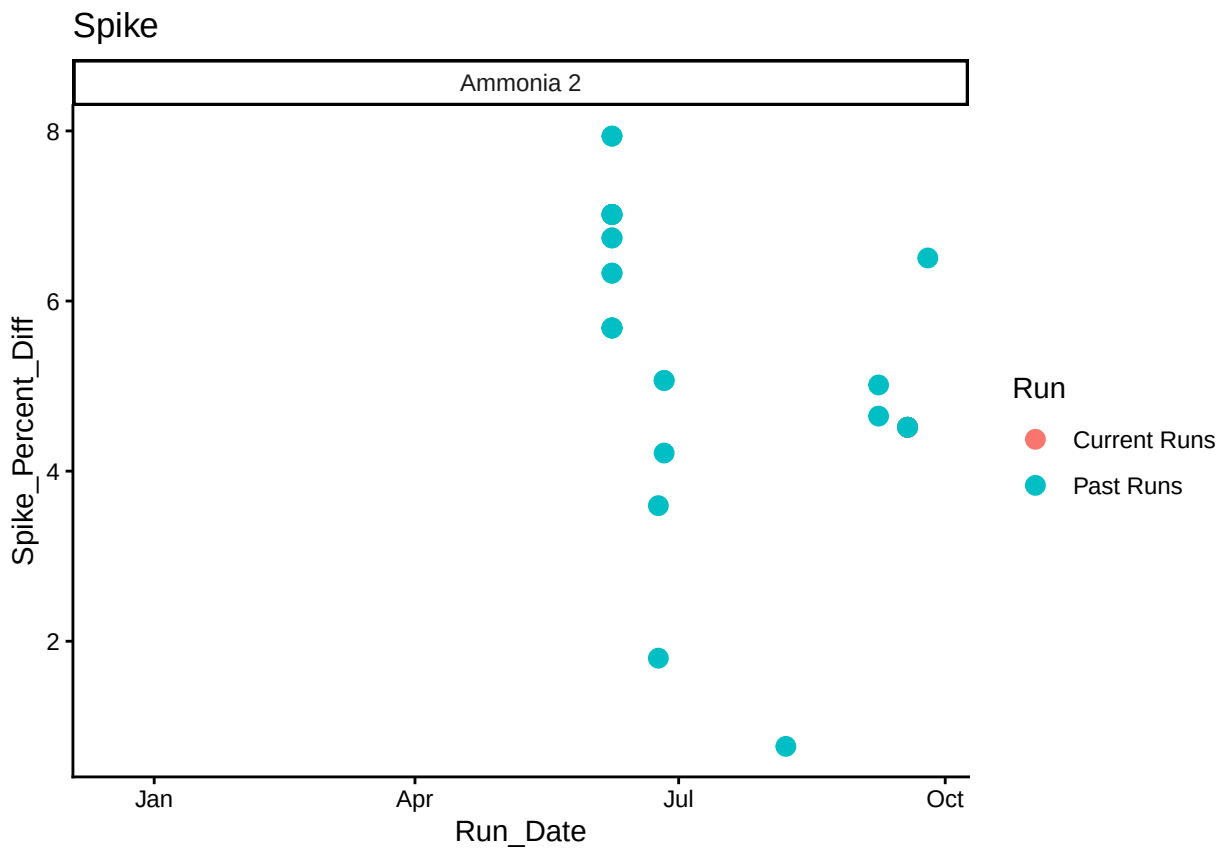
##Add QAQC Flags

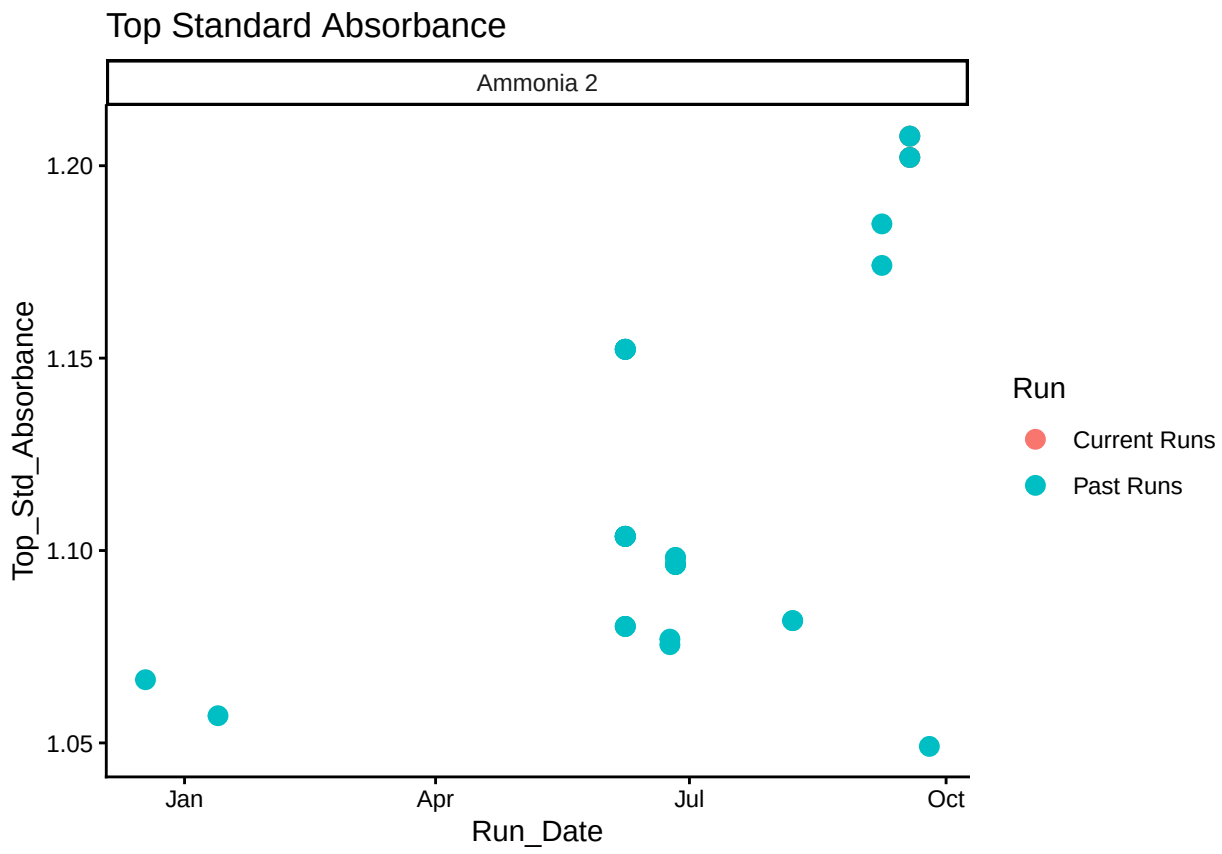
0.10 Create Log for QAQC

```
## Warning in left_join(Top_Standards, PE_Check, by = join_by(Test, Run_Date)): Detected an unexpected r
## i Row 1 of ‘x‘ matches multiple rows in ‘y‘.
## i Row 1 of ‘y‘ matches multiple rows in ‘x‘.
## i If a many-to-many relationship is expected, set ‘relationship =
##   "many-to-many"‘ to silence this warning.

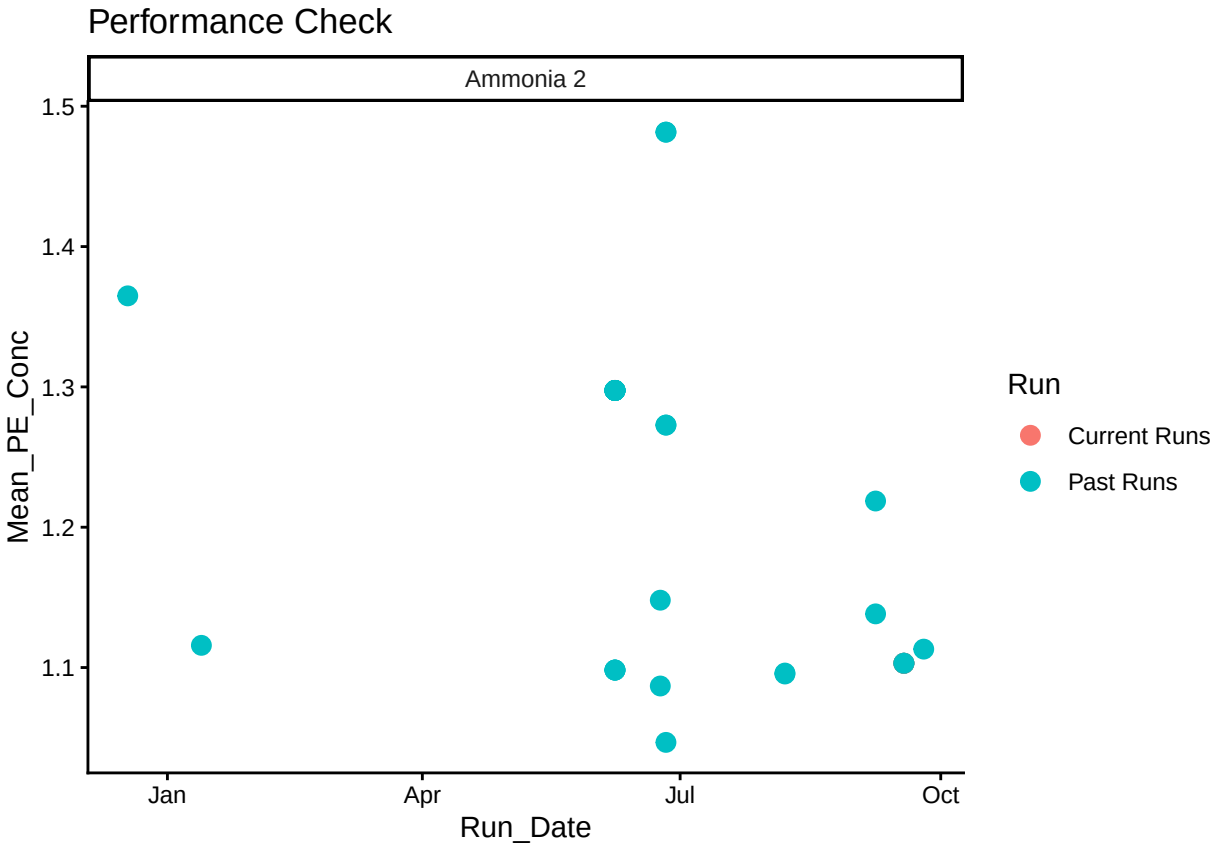
## ‘summarise()‘ has grouped output by ’Test’, ’Run_Date’, ’Top_Std_Absorbance’,
## ’Mean_PE_Conc’. You can override using the ‘.groups‘ argument.

## Warning: Removed 6 rows containing missing values or values outside the scale range
## (‘geom_point()‘).
```


Spike
Ammonia 2
Spike_Percent_Diff
8
6
4
2
Jan
Apr
Jul
Oct
Run_Date
Run
Current Runs
Past Runs




```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



##Write QAQC Log

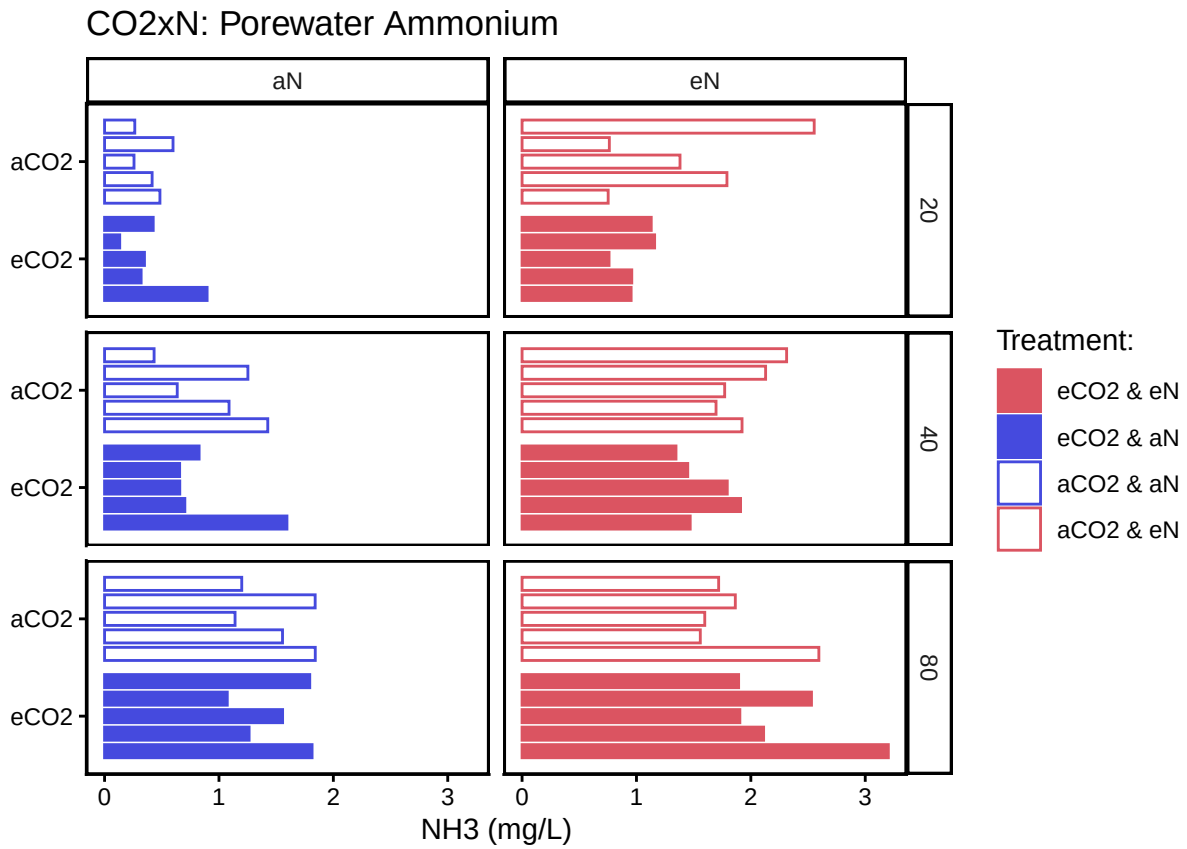
0.11 Merge samples with Metadata & check all samples are present

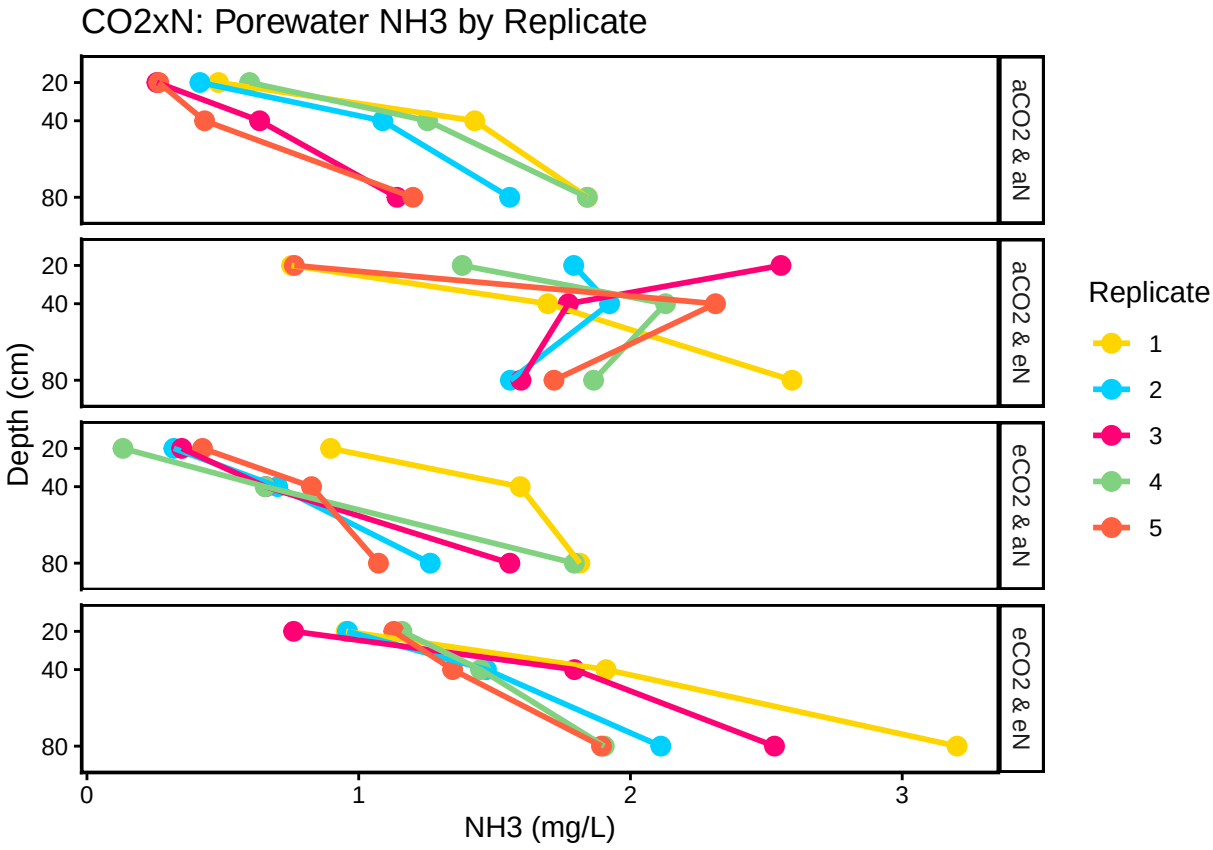
```
## Merge Samples with Metadata

## All sample IDs are present in data
```

0.12 Organize Data

0.13 Visualize Data





0.14 Export Processed Data

#end